

$$\begin{aligned}
\mathbf{g}_1 \rightarrow \mathbf{g}_1 + \frac{1}{16\pi^2} & \left(\frac{19}{12} \mathbf{g}_1^3 - \frac{1}{6} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\tilde{\mathbf{d}}}^{\mathbf{p}}] + \frac{1}{9} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\tilde{\mathbf{d}}}^{\mathbf{p}}] - \right. \\
& \frac{1}{2} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\tilde{\mathbf{e}}}^{\mathbf{p}}] + \frac{1}{3} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\tilde{\mathbf{e}}}^{\mathbf{p}}] - \frac{1}{4} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\tilde{\mathbf{l}}}^{\mathbf{p}}] + \frac{1}{6} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\tilde{\mathbf{l}}}^{\mathbf{p}}] - \\
& \frac{1}{12} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\tilde{\mathbf{q}}}^{\mathbf{p}}] + \frac{1}{18} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\tilde{\mathbf{q}}}^{\mathbf{p}}] - \frac{2}{3} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\tilde{\mathbf{u}}}^{\mathbf{p}}] + \\
& \left. \frac{4}{9} \sum_{\mathbf{p}} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\tilde{\mathbf{u}}}^{\mathbf{p}}] - \frac{1}{4} \mathbf{g}_1^3 \text{LF}_{2,0}[\mathbf{m}_{\oplus}] + \frac{1}{6} \mathbf{g}_1^3 \text{LF}_{3,-1}[\mathbf{m}_{\oplus}] - \frac{1}{3} \mathbf{g}_1^3 \text{LF}_{3,-1}[\tilde{\mu}] \right)
\end{aligned}$$